

Project Architecture

ANSYS APDL
simple_beam.inp



ansys_parser.py
build_native_model()



post.py
solve() + Post (3D viz)



beam_element.py
12x12 + section properties



numpy_ebe/
ebe_pcg.py
(serial, validation)



jax_ebe/
ebe_pcg.py
(vmap GPU parallel)

-> Benchmark & Comparison

BEAM4